Education

Bachelor of Technology (B.Tech) — Major in Electronics, Minor in Computer Science Engineering
CGPA: 7

Senior Secondary (Class XII – Science): 71%
Secondary (Class X): 72.6%

Work Experience

Machine Learning Research Intern
Jun 2025 – Jul 2025

- Conducted an extensive literature review of **50+ research papers** on medical image analysis and deep learning for leukemia detection to guide model selection and experimental design.
- Developed **CNN-based classification models** (VGG16, ResNet50, InceptionV3) for leukemia image classification.
- Built **end-to-end ML pipelines** for data preprocessing, feature extraction, model training, and evaluation using TensorFlow and PyTorch.

Data Scientist Intern
May 2024 – Oct 2024

- Analyzed large-scale datasets using **Python, Excel, and Power BI** to support decisions across infrastructure and transportation projects.
- Built forecasting dashboards optimizing **material and cost planning**, influencing tender bidding strategies.
- Automated reporting pipelines, reducing manual work by **35%** and improving analytical accuracy.

Project Work

GenAI Project – RAG-Based Research Paper Assistant

Developed a Generative AI system leveraging **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)** for document-grounded question answering over academic research papers.

The system integrates semantic document retrieval using dense vector embeddings and similarity search with controlled LLM-based generation, ensuring responses remain grounded in source documents. By separating retrieval and generation, the system reduces hallucination and improves factual consistency in scientific document analysis.

Tech:
Python, RAG, Sentence-BERT, FAISS, HuggingFace Transformers, LangChain, NLP, Semantic Search

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InViTNet — Hybrid Transformer–Convolution Vision-Language Model (Ongoing)

Built a multi-branch medical imaging model combining **Vision Transformers (ViT), CNN, RNN, and a Vision-Language model** to fuse global attention, spatial features, sequential patterns, and multimodal image-text understanding.

Enhanced classification accuracy on **BloodMNIST** using cross-modal embeddings and transformer-based reasoning.

Tech:
Python, PyTorch, ViT, CNN, RNN, Vision-Language Models

ChurnStarPRED – Predictive AI for Customer Churn

Developed an explainable AI system to predict customer churn and identify underlying reasons from **unstructured customer emails**.

- Applied NLP techniques (sentiment, intent detection, complaint extraction) combined with **XGBoost** for churn prediction.
- Implemented **PII redaction** and explainable outputs to generate human-readable churn reasons and recommended retention actions.

Tech Stack:
Python, NLP (TextBlob), XGBoost, Scikit-learn, Pandas, NumPy, Explainable AI, PII Redaction, Gradio

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Technical Skills

Programming Languages:
Python, C/C++, SQL

Frameworks & Libraries:
TensorFlow, PyTorch, Scikit-Learn, Keras, OpenCV, Transformers

ML & AI Expertise:
Supervised Learning, Unsupervised Learning, CNNs, NLP, MLOps, Deep Learning, Model Deployment

Data Analytics & Visualization:
Pandas, NumPy, Matplotlib, Power BI, Tableau

Cloud & Tools:
AWS, Google Cloud, Docker, Gradio, Git, MySQL, Jupyter, VS Code

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Certifications and Achievements

Achievement

- Ranked **53rd in the Amazon ML Challenge 2025** among thousands of participants.

Certifications

- Data Analytics Virtual Internship – Forensic Technology
- Introduction to Data Science Job Simulation
- AWS Solution Architecture Virtual Simulation

Leadership & Responsibilities

Member – Technical Student Forum

- Contributed to planning, coordination, and management of institutional events.

Member – Professional Management Association

- Managed the official online portal, coordinated tasks, and supported internal communications.